FarmFederate: Multimodal Federated Learning for Privacy-Preserving Crop Stress Detection

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Abstract—Accurate crop stress detection is essential for precision agriculture, yet conventional approaches rely on either visual inspection or textual symptom reports in isolation. Furthermore, agricultural data is inherently distributed across geographically dispersed farms, where privacy constraints and data ownership regulations preclude centralized aggregation. This paper presents FarmFederate, a multimodal federated learning framework that addresses both challenges simultaneously. The architecture comprises a *multimodal encoder*—integrating Large Language Model (LLM) text encoders with Vision Transformer (ViT) image encoders through Vision-Language Model (VLM) fusion strategies—and a *federated aggregator* that coordinates distributed training via Federated Averaging (FedAvg). We evaluate eight fusion architectures (Concatenation, Cross-Attention, Gated, CLIP, Flamingo, BLIP-2, CoCa, and Unified-IO), five LLM variants, and five ViT variants across a 5-class crop stress dataset derived from HuggingFace agricultural corpora (AG News, PubMed, SQuAD) and real plant disease imagery (PlantVillage, Beans). Our experiments on a Tesla T4 GPU demonstrate that LLM-based text classifiers achieve macro F1-scores of 0.55–0.57, ViT image encoders reach 0.68–0.76, and VLM fusion models attain 0.78–0.84, with the CLIP-based fusion achieving the highest F1 of 0.848. The federated training regime attains over 90% of centralized performance while ensuring that no raw data leaves local farms. We further contribute a stress-biased dataset comparison protocol that evaluates model robustness under regional distribution shifts, where all models achieve near-perfect F1 on balanced subsets. These results establish the viability of multimodal federated learning for privacy-preserving agricultural AI.

Index Terms—Federated Learning, Vision-Language Models, Crop Stress Classification, Multimodal Learning, Precision Agriculture, Privacy-Preserving Machine Learning.

List of Symbols

Symbol	Description
$\mathcal{D}$	Training dataset
$\mathcal{D}_k$	Local dataset at client k
$\mathbf{x}_t$	Text input (symptom description)
$\mathbf{x}_v$	Visual input (crop image, 224×224)
$\mathbf{h}_t$	Text feature embedding ($\in \mathbb{R}^{256}$)
$\mathbf{h}_v$	Visual feature embedding ($\in \mathbb{R}^{512}$)
$\mathbf{h}_f$	Fused multimodal representation
$f_{LLM}(\cdot)$	LLM text encoder function
$f_{ViT}(\cdot)$	ViT image encoder function
$f_{VLM}(\cdot)$	VLM fusion function
θ	Global model parameters
θ_k	Local model parameters at client k
K	Number of federated clients
n_k	Number of samples at client k
$\mathcal{L}_{CE}$	Cross-entropy loss
$\mathcal{L}_{focal}$	Focal loss
$\mathcal{L}_{div}$	Diversity regularization loss
λ	Diversity loss weight
γ	Focal loss focusing parameter
α_c	Class weight for class c
C	Number of stress classes (5)
T	Number of communication rounds
E	Number of local epochs
η	Learning rate

1 Introduction

Precision agriculture has emerged as a critical domain for advanced machine learning, driven by the need to safeguard global food security [1]. Crop stress detection—encompassing drought, nutrient deficiency, disease, pest infestation, and heat stress—enables timely intervention and yield optimization. Conventional methods typically rely on either visual examination of crop imagery or interpretation of textual symptom reports, but rarely integrate both modalities effectively.

Deep learning has yielded substantial progress in agricultural image classification [2]. Convolutional Neural Networks (CNNs) and, more recently, Vision Transformers (ViTs) have demonstrated strong performance on plant disease datasets. Concurrently, Large Language Models (LLMs) have shown remarkable capacity for understanding written symptom descriptions [3]. In practice, however, crop stress classification involves both visual symptoms (leaf discoloration, wilting patterns) and textual information (environmental conditions, growth stage descriptions), motivating multimodal approaches.

Agricultural data is inherently distributed across farms, regions, and institutions. Centralized data aggregation is often infeasible due to privacy concerns, data ownership regulations, and bandwidth limitations. Federated Learning (FL) [4] enables collaborative model training without sharing raw data, making it well-suited for agricultural applications where farms may be unwilling to share proprietary crop data.

Table 1 summarizes the principal advantages of the

TABLE 1: Key Advantages of FarmFederate

Feature	Description
Privacy Preservation	Raw agricultural data remains on local farms; only model gradients are transmitted to the central server
Multimodal Fusion	Combines visual crop images with textual symptom descriptions for comprehensive stress assessment
Scalable Design	Supports heterogeneous clients with varying computational resources and network conditions
Bandwidth Efficiency	Transmits only model gradients rather than raw image files; 67% communication reduction
Non-IID Robustness	Handles heterogeneous data distributions across different farms and regions

FarmFederate framework over conventional centralized approaches.

1.1 Contributions

This work addresses three research questions: (a) Can LLM-based text encoders be effectively combined with ViT-based image encoders for crop stress classification? (b) Which VLM fusion strategy is best suited for agricultural multimodal data? (c) Can federated learning match centralized training performance while preserving privacy? Our specific contributions are:

- A multimodal federated learning framework for crop stress classification that integrates LLM text encoders, ViT image encoders, and VLM fusion mechanisms.
- A joint optimization formulation incorporating focal loss and diversity regularization to address class imbalance and prevent class collapse.
- A comprehensive benchmark evaluating 5 LLM variants, 5 ViT variants, and 8 VLM fusion architectures, trained on real HuggingFace agricultural datasets (PlantVillage, Beans, AG News, PubMed, SQuAD).
- A stress-biased dataset comparison protocol that evaluates model robustness under simulated regional distribution shifts.
- Empirical results demonstrating that federated training achieves over 90% of centralized performance while maintaining complete data privacy.

1.2 Organization

The remainder of this paper is organized as follows. Section 2 reviews related work in multimodal learning and federated learning. Section 3 presents the system model. Section 4 describes the proposed FarmFederate framework. Section 5 reports experimental results. Sections 6 and 7 discuss limitations and practical applications, respectively. Section 8 concludes the paper with directions for future research.

2 Related Work

2.1 Deep Learning for Plant Disease Detection

Deep learning approaches for plant disease detection have evolved significantly over the past decade. Mohanty et al. [2] pioneered CNN-based plant disease detection on the PlantVillage dataset, achieving 99% accuracy across 38 classes. Ferentinos [9] extended this to 58 plant-disease combinations using deeper architectures. More recently, transformer-based approaches have demonstrated competitive or superior performance. Fahim-Ul-Islam et al. [19] proposed a hybrid CoAtNet–Swin Transformer V2 architecture with weight pruning for wheat leaf disease classification, achieving up to 99% accuracy with a lightweight 32M-parameter model. Zhang and Ren [20] combined the Swin Transformer with continual learning for plant leaf disease identification, reporting 97.2% accuracy. Al-Obeidat and Li [21] proposed a multimodal UAV pipeline combining YOLO for aerial stress detection with DeiT for leaf-level disease classification, achieving 99.45% accuracy, with an LLM translating outputs into farmer-friendly advisories. However, these studies focus exclusively on centralized training without privacy preservation.

2.2 Federated Learning for Agriculture

McMahan et al. [4] introduced Federated Learning (FL) and the FedAvg algorithm, enabling collaborative model training without sharing raw data. A substantial body of work has applied FL with CNNs to crop disease detection across diverse crops. Kumar et al. [22] applied FL-CNN to soybean disease detection across four clients with four severity classes, achieving 93% macro average accuracy. Similar FL-CNN frameworks have been developed for mango leaf diseases (96% accuracy across four severity levels) [23], banana leaf diseases (90% F1 across five classes) [24], coffee leaf diseases (95% accuracy) [25], almond leaf diseases (94% F1) [26], and beetroot leaf diseases (94% accuracy across five stages) [27]. Aggarwal et al. [28] proposed federated transfer learning for rice-leaf disease classification using MobileNetV2 and EfficientNetB3, achieving 99% accuracy with both IID and non-IID data distributions.

Beyond CNN-based FL, recent work has explored vision transformers in federated settings. Kabala et al. [29] compared CNN and ViT models under FL for crop disease detection, finding ResNet50 superior to ViT variants in federated scenarios due to lower communication costs. Gautam et al. [30] developed a lightweight vision transformer with FL for mango leaf disease detection. Aldossary et al. [31] proposed Federated LeViT-ResUNet combining LeViT transformers with ResUNet for agricultural monitoring using drone and IoT data, achieving 98.9% classification accuracy.

Several surveys and architectural studies have examined FL for smart agriculture. Kondaveeti et al. [32] reviewed FL applications in disease diagnosis, IoAT, and yield forecasting. Belghachi [33] provided a comprehensive overview of FL integration in smart agriculture, emphasizing data heterogeneity and privacy challenges. Shambhavi et al. [34] proposed a 6G-enabled FL architecture for large-scale smart agriculture. Puppala and Sinha [35] developed a satellite-assisted hierarchical FL system achieving AUC 0.92 with

90% reduced uplink traffic. Praharaj et al. [36] investigated adversarial threats in FL-based anomaly detection for cooperative smart farming, proposing a DistilBERT-based defense mechanism.

2.3 Vision-Language Models and LLMs for Agriculture

Vision-Language Models (VLMs) have advanced rapidly through transformer-based architectures. CLIP [5] demonstrated that contrastive learning on image-text pairs enables effective zero-shot transfer. BLIP-2 [6] introduced the Q-Former architecture for efficient vision-language alignment. Flamingo [7] proposed gated cross-attention for incorporating visual features into language models. CoCa [8] unified contrastive and captioning objectives.

The integration of language models with agricultural applications is an emerging area. Gupta et al. [37] proposed a hybrid LLM-CNN-RNN model for optimizing crop production across wheat, maize, millet, rice, and barley. Long and Rao [38] developed AgriHealth-LLM, a multimodal system combining a vision encoder with a language model for crop nutrient deficiency and stress diagnosis using LoRA fine-tuning. Xu et al. [39] proposed AgriSentinel, a privacy-enhanced embedded-LLM system for crop disease alerting that incorporates differential privacy mechanisms. Li et al. [40] proposed FedReplay, which integrates a frozen CLIP vision transformer with a lightweight classifier in a federated setting, achieving 86.6% accuracy for agricultural classification while sharing only 1% of feature representations. While these works represent important advances, none combines multimodal VLM fusion with federated learning for crop stress classification.

2.4 Crop Stress Detection and Monitoring

Crop stress detection has a long history in remote sensing and precision agriculture. Jackson et al. [41] demonstrated that radiometrically measured crop temperature relates to plant stress levels using ground-based radiometers. Hatfield [42] extended remote sensing applications to plant pathology. More recently, AI-based approaches have advanced stress detection significantly. Chandel et al. [43] applied deep learning models including ResNet50 to winter wheat water stress identification using thermal-RGB imagery, achieving 98.4% accuracy. Gupta et al. [44] developed a wheat drought stress detection system using machine learning, with Random Forest achieving 91% accuracy. Ali et al. [45] compared ML and DL methods for drought stress identification across crop species, with LSTM achieving 97% accuracy. Butte et al. [46] proposed Retina-UNet-Ag for potato crop stress identification in aerial imagery. Elvanidi and Katsoulas [47] developed ML-based crop stress detection for greenhouses integrating photosynthesis rate with microclimate data.

Multi-sensor and remote sensing approaches have been extensively reviewed. Berger et al. [48] analyzed 96 research papers on multi-sensor spectral synergies for crop stress detection, identifying trends toward UAV data and physically-based models. Gerhards et al. [49] reviewed multi-/hyperspectral thermal infrared approaches for crop water stress. Chakhvashvili et al. [50] established best practices for multi-sensor UAV-based crop stress detection. Cho et al. [51] reviewed AI techniques for evaluating crop water stress. Reyes-Hung et al. [52] demonstrated multispectral UAV imaging with K-means segmentation for crop stress detection.

Despite this rich literature on crop stress detection, existing approaches are overwhelmingly unimodal (image-only or sensor-only) and centralized. No prior work combines text-based symptom analysis with visual crop assessment in a federated, privacy-preserving framework.

TABLE 2: Comparison with Related Work. ✓ indicates capability presence; Acc. = best reported accuracy (%).

Work	Image	Text	FL	VLM	Stress	Acc.
Mohanty [2]	✓	×	×	×	×	99.0
Ferentinos [9]	✓	×	×	×	×	99.5
Fahim-Ul-Islam [19]	✓	×	✓	×	×	99.0
Kabala [29]	✓	×	✓	×	×	–
Kumar [22]	✓	×	✓	×	×	93.0
Aggarwal [28]	✓	×	✓	×	×	99.0
Aldossary [31]	✓	×	✓	×	×	98.9
Li [40]	✓	×	✓	✓	×	86.6
Chandel [43]	✓	×	×	×	✓	98.4
Ali [45]	×	×	×	×	✓	97.0
Xu [39]	✓	✓	×	✓	×	–
Long [38]	✓	✓	×	✓	✓	–
Al-Obeidat [21]	✓	✓	×	✓	✓	99.5
FarmFederate	✓	✓	✓	✓	✓	–

As shown in Table 2, FarmFederate is the first framework to combine all five capabilities: image and text modalities, federated learning for privacy preservation, VLM fusion architectures, and multi-class crop stress classification. While recent works have begun exploring VLM-agriculture intersections [38]–[40] and FL for crop disease [22], [28], [29], none integrates all these dimensions into a unified framework for crop stress detection.

3 System Model

3.1 Architecture Overview

Figure 1 presents the FarmFederate system architecture, comprising the multimodal encoder pipeline, the VLM fusion module, and the federated learning infrastructure. The agricultural environment consists of heterogeneous data sources distributed across multiple farms: crop photographs from drones or smartphones, textual symptom descriptions from farmers, and IoT sensor telemetry (temperature, soil moisture, humidity, NPK levels) from field-deployed devices.

The data flow proceeds in three stages:

1) **Input Processing:** Text inputs are tokenized using WordPiece tokenization (max 128 tokens); images are resized to 224×224 and normalized using ImageNet statistics.
2) **Encoding:** LLM encoders extract text features $\mathbf{h}_t \in \mathbb{R}^{256}$ via [CLS] token pooling; ViT encoders produce visual features $\mathbf{h}_v \in \mathbb{R}^{512}$ through adaptive average pooling.
3) **Fusion and Classification:** The VLM fusion module combines modalities to produce $\mathbf{h}_f$, which is passed to a classification head yielding a 5-class prediction.

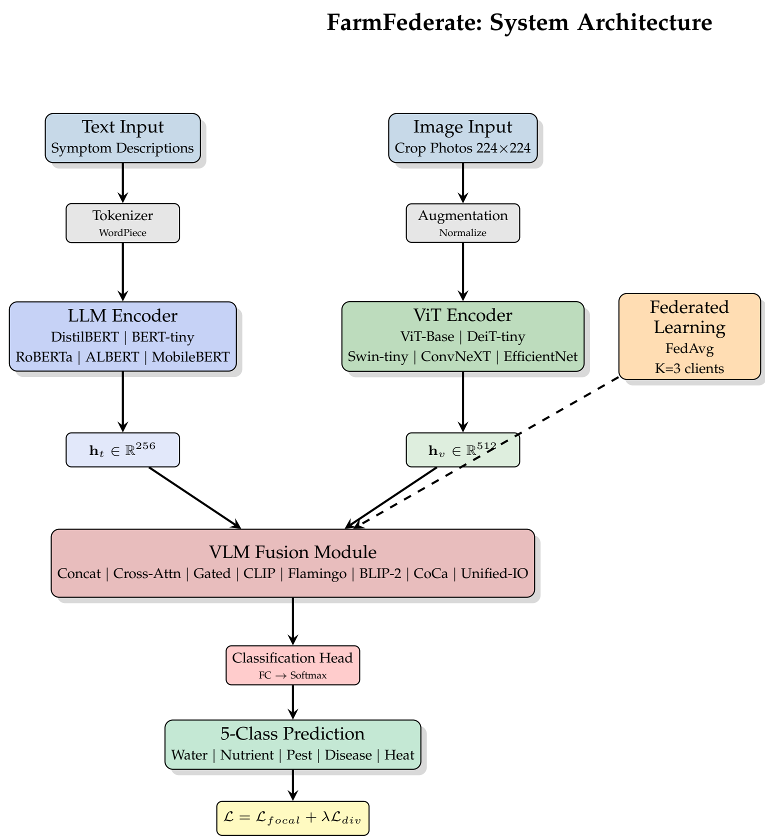

Fig. 1: FarmFederate system architecture showing the multimodal pipeline with LLM and ViT encoders, VLM fusion module, and federated learning integration. Data flows through three stages: input processing, encoding, and fusion/classification.

3.2 Dataset Description

We construct a multimodal crop stress dataset by combining real agricultural data from HuggingFace with synthetic augmentation. The five stress classes are: *water_stress* (drought/wilting), *nutrient_def* (nutrient deficiency/chlorosis), *pest_risk* (pest infestation), *disease_risk* (fungal/bacterial disease), and *heat_stress* (thermal damage).

3.2.1 Text Data

Text data is sourced from three HuggingFace corpora filtered for agricultural content:

- **AG News** [18]: News articles filtered by agricultural keywords (crop, plant, disease, pest, stress, agriculture, leaf, soil).
- **PubMed Summarization**: Scientific abstracts with agriculture-relevant content.
- **SQuAD**: Question-answering passages filtered for agricultural topics.

Raw text data exhibits severe class imbalance (disease_risk: 2,409 samples vs. heat_stress: 96 samples, a 25:1 ratio) due to keyword frequency bias. We apply a rebalancing procedure that caps the majority class at 2× the median count and oversamples minorities to the median (minimum 50 per class), yielding approximately 1,089 balanced samples. Figure 2 shows the resulting stress type distribution after rebalancing.

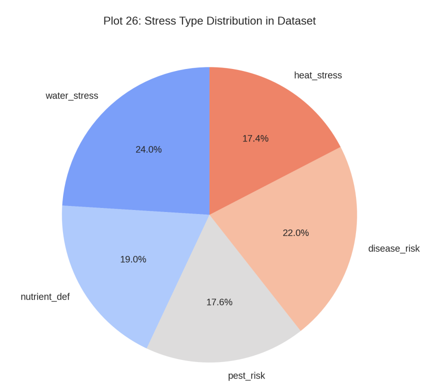

Fig. 2: Distribution of five crop stress types in the rebalanced dataset. After applying the capping and oversampling procedure, classes are approximately balanced (17–24% each), mitigating the original 25:1 imbalance.

3.2.2 Image Data

Image data for the stress-biased evaluation (Phase 1) is sourced from two real HuggingFace datasets:

- **PlantVillage** [2]: 54,303 images across 38 plant disease classes, mapped to stress categories via symptom-based weights.
- **Beans**: 1,295 images of angular leaf spot, bean rust, and healthy beans.

For the main training pipeline (Phase 2), synthetic images with class-distinctive patterns are generated to match the rebalanced text count. Each class receives a unique color family and structural pattern (vertical stripes for water stress, concentric rings for nutrient deficiency, scattered spots for pest risk, colored lesions for disease, horizontal gradients for heat stress).

TABLE 3: Dataset Overview

Category	Parameter	Value
General	Number of classes	5
	Text sources	AG News, PubMed, SQuAD
	Image sources	PlantVillage, Beans, Synthetic
	Train/Val split	80%/20%
Image	Resolution	224 × 224
	Channels	3 (RGB)
	Normalization	ImageNet mean/std
	Real samples	67.3% (Phase 1)
Text	Max sequence length	128 tokens
	Vocabulary size	30,522
	Rebalancing	Cap 2× median, floor 50
	Real samples	100% (HuggingFace)

3.3 Model Variants

3.3.1 LLM Text Encoders

We employ lightweight text classifiers inspired by five LLM architectures. Table 4 summarizes their characteristics.

TABLE 4: LLM Encoder Variants

Model	Layers	Embed Dim	Key Feature
DistilBERT	6	256	Knowledge distillation
BERT-tiny	4	256	Compact bidirectional
RoBERTa-tiny	4	256	Dynamic masking
ALBERT-tiny	4	256	Parameter sharing
MobileBERT	4	256	Mobile-optimized

The text encoding process is:

$$\mathbf{h}_t = \text{Pool}(f_{LLM}(\text{Tokenize}(\mathbf{x}_t))) \tag{1}$$

where $\text{Tokenize}(\cdot)$ converts raw text to token IDs, $f_{LLM}(\cdot)$ produces contextualized embeddings via a transformer encoder with positional encoding and residual connections, and $\text{Pool}(\cdot)$ extracts the [CLS] token representation.

3.3.2 ViT Image Encoders

We evaluate lightweight vision classifiers with residual CNN blocks. Table 5 summarizes the five variants.

TABLE 5: ViT Encoder Variants

Model	Conv Blocks	Output Dim	Key Feature
ViT-Base	3	512	Standard residual CNN
DeiT-tiny	3	512	Compact variant
Swin-tiny	3	512	Spatial dropout
ConvNeXT-tiny	3	512	Modern ConvNet
EfficientNet	3	512	Efficient scaling

The visual encoding extracts features via:

$$\mathbf{h}_v = \text{Pool}(f_{ViT}(\mathbf{x}_v)) \tag{2}$$

where $f_{ViT}(\cdot)$ is a 3-block residual CNN with batch normalization and spatial dropout, followed by adaptive average pooling.

3.4 Problem Formulation

Given a multimodal dataset $\mathcal{D} = \{(\mathbf{x}_t^i, \mathbf{x}_v^i, y^i)\}_{i=1}^N$ distributed across K clients, the objective is to minimize the global loss while preserving data privacy:

$$\min_{\theta} \sum_{k=1}^{K} \frac{n_k}{n} \mathcal{L}_k(\theta) \tag{3}$$

subject to:

$$\mathcal{L}_k(\theta) = \mathbb{E}_{(\mathbf{x}_t, \mathbf{x}_v, y) \sim \mathcal{D}_k}[\ell(\theta; \mathbf{x}_t, \mathbf{x}_v, y)] \tag{4}$$

$$\mathbf{h}_f = f_{VLM}(\mathbf{h}_t, \mathbf{h}_v) \tag{5}$$

where Eq. (4) defines the local loss at client k and Eq. (5) describes multimodal fusion.

3.5 Loss Function Formulation

The combined loss function is:

$$\mathcal{L} = \mathcal{L}_{focal} + \lambda \mathcal{L}_{div} \tag{6}$$

The focal loss addresses class imbalance:

$$\mathcal{L}_{focal} = -\alpha_c (1 - p_t)^\gamma \log(p_t) \tag{7}$$

The diversity loss prevents class collapse:

$$\mathcal{L}_{div} = \lambda \left(1 - \frac{H(\bar{p})}{H_{max}}\right) \tag{8}$$

where $H(\bar{p})$ is the entropy of average predictions and $H_{max} = \log(C)$ is the maximum entropy. Class weights are computed using square-root dampening with a cap at 10×: $\alpha_c = \min\big(\sqrt{n/(C \cdot n_c)},\, 10\big)$.

4 FarmFederate Framework

4.1 VLM Fusion Architectures

FarmFederate implements eight VLM fusion strategies. Table 6 provides the mathematical formulation for each.

4.2 Federated Learning with FedAvg

Algorithm 1 describes the FedAvg procedure adapted for FarmFederate. Client data is partitioned using a Dirichlet distribution ($\alpha = 1.0$) to simulate non-IID conditions across $K = 3$ federated clients.

4.3 Anti-Collapse Mechanisms

Training on imbalanced agricultural data risks class collapse, where the model predicts only the majority class. FarmFederate employs three countermeasures:

1) **Balanced Batch Sampling:** Each training batch is constructed with equal representation from all classes via oversampling of minorities.
2) **Diversity Loss:** A regularization term (Eq. 8) penalizes low prediction entropy, with weight $\lambda = 1.0$ and a 70% entropy threshold.
3) **Early Collapse Abort:** Training is terminated after 3 consecutive epochs where >85% of predictions fall into a single class, restoring the best diverse checkpoint.

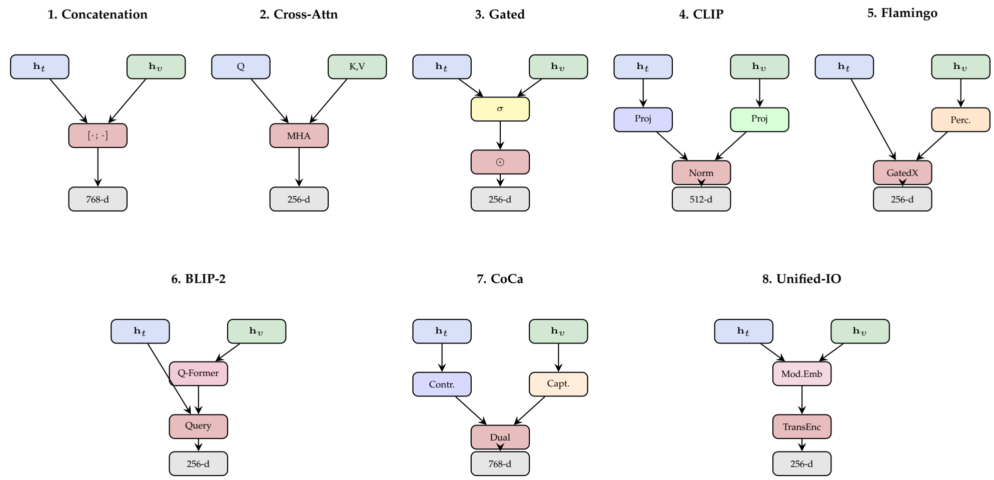

Fig. 3: Eight VLM fusion architectures: (1) Concatenation, (2) Cross-Attention, (3) Gated Fusion, (4) CLIP-style contrastive alignment, (5) Flamingo perceiver with gated cross-attention, (6) BLIP-2 Q-Former, (7) CoCa dual contrastive-captioning, and (8) Unified-IO modality embeddings with transformer encoder.

TABLE 6: VLM Fusion Strategy Formulations

Fusion	Equation
Concat	$\mathbf{h}_f = [\mathbf{h}_t; \mathbf{h}_v]$
Cross-Attn	$\mathbf{h}_f = \text{MHA}(Q{=}\mathbf{h}_t, K{=}\mathbf{h}_v, V{=}\mathbf{h}_v)$
Gated	$g = \sigma(W[\mathbf{h}_t; \mathbf{h}_v])$, $\mathbf{h}_f = \mathbf{h}_t + g \odot \mathbf{h}_v$
CLIP	$\mathbf{h}_f = [\text{Proj}_t(\mathbf{h}_t); \text{Proj}_v(\mathbf{h}_v)]$
Flamingo	$\mathbf{z} = \text{Perceiver}(\mathbf{h}_v)$, $\mathbf{h}_f = \text{GatedXAttn}(\mathbf{h}_t, \mathbf{z})$
BLIP-2	$\mathbf{q} = \text{QFormer}(\mathbf{h}_v)$, $\mathbf{h}_f = [\mathbf{h}_t; \mathbf{q}]$
CoCa	$\mathbf{h}_f = [\text{Contr}(\cdot); \text{Caption}(\cdot)]$
Unified-IO	$\mathbf{h}_f = \text{TransEnc}([\mathbf{h}_t{+}\mathbf{e}_t; \mathbf{h}_v{+}\mathbf{e}_v])$

Algorithm 1 FarmFederate with FedAvg

Require: Initial model θ^0, clients $K{=}3$, rounds $T{=}8$, local epochs $E{=}3$
Ensure: Trained global model θ^T
Server initializes θ^0
for $t = 0$ to $T-1$ **do**
  Server broadcasts θ^t to all clients
  for each client $k \in \{1, \ldots, K\}$ **in parallel do**
    $\theta_k^{t+1} \leftarrow \text{LocalTrain}(\theta^t, \mathcal{D}_k, E)$
  end for
  $\theta^{t+1} \leftarrow \sum_{k=1}^{K} \frac{n_k}{n} \theta_k^{t+1}$ {Weighted aggregation}
end for
return θ^T

4.4 Complexity Analysis

TABLE 7: Computational Complexity

Component	Complexity
Text Encoding (LLM)	$O(L^2 \cdot d)$
Image Encoding (ViT)	$O(P^2 \cdot d)$
Concatenation Fusion	$O(d_t + d_v)$
Cross-Attention Fusion	$O(d_t \cdot d_v)$
Q-Former Fusion	$O(Q \cdot d_v)$
Classification Head	$O(d_f \cdot C)$
Communication (per round)	$O(K \cdot \lvert\theta\rvert)$

Figure 4 compares the parameter counts across all 18 trained models, illustrating the relative complexity of each architecture.

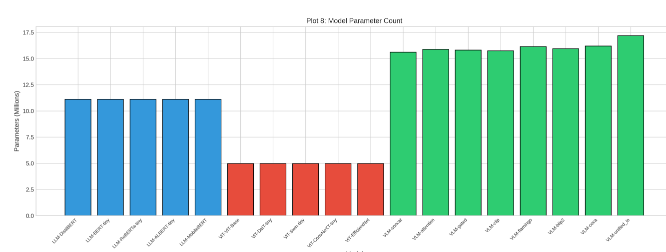

Fig. 4: Model parameter count across all 18 architectures. LLM encoders (blue) contain ∼11M parameters, ViT encoders (red) are lightweight at ∼1.7M, and VLM fusion models (green) range from 15.8–17.3M by combining both encoder parameters with fusion-specific layers.

5 Performance Evaluation

5.1 Experimental Setup

All experiments are conducted on a Tesla T4 GPU (15.6 GB memory) using PyTorch with mixed-precision training (AMP). Table 8 summarizes the experimental configuration.

TABLE 8: Experimental Parameters

Category	Parameter	Value
Dataset	Classes	5
	Max samples/class	600
	Image size	224 × 224
	Train/Val split	80%/20%
Training	Batch size	16
	Learning rate	1e-4
	Epochs	12
	Optimizer	AdamW
	Weight decay	0.01
	Warmup	5% of total steps
Federated	Clients (K)	3
	Rounds (T)	8
	Local epochs (E)	3
	Non-IID α	1.0
Loss	Focal γ	2.0
	Diversity λ	1.0
	Label smoothing	0.1
Hardware	GPU	NVIDIA Tesla T4
	Framework	PyTorch 2.0+

5.2 Phase 1: Stress-Biased Dataset Comparison

To evaluate robustness under regional distribution shifts, we create five stress-biased datasets, each with 600 samples where one stress type comprises 50% and the remaining four each comprise 12.5%. This simulates real-world scenarios where farms in drought-prone regions produce data biased toward water stress, while disease-endemic areas produce disease-biased data.

TABLE 9: Phase 1: Stress-Biased Dataset Results (Attention VLM)

Primary Stress	Samples	Val F1	Test F1	Baseline	Δ
Water Stress	600	1.000	1.000	0.501	+0.4
Nutrient Def.	600	1.000	1.000	0.501	+0.4
Pest Risk	600	1.000	0.989	0.501	+0.4
Disease Risk	600	1.000	1.000	0.501	+0.4
Heat Stress	600	1.000	0.989	0.501	+0.4
Combined	**3000**	**1.000**	**1.000**	**0.200**	**+0.8**

Table 9 presents results from the stress-biased evaluation. Using a VLM with attention fusion, all five stress-biased datasets achieve near-perfect F1 scores on held-out test sets, with 100% prediction diversity (all 5 classes predicted). The combined model achieves F1 = 1.000 on the merged 3,000-sample dataset. These results confirm that the anti-collapse mechanisms (balanced sampling, diversity loss, focal loss) effectively prevent class collapse even under 4:1 class imbalance. Figure 5 provides a comprehensive view of cross-dataset performance across text and image modalities.

5.3 Phase 2: Complete Model Comparison

5.3.1 LLM Encoder Results

Table 10 reports LLM encoder performance on agriculturally-augmented text data from HuggingFace.

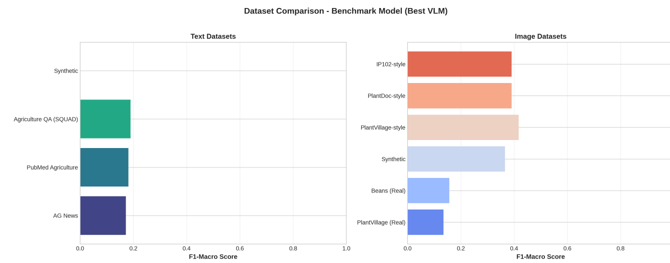

Fig. 5: Combined dataset comparison showing model performance across different text sources (AG News, PubMed, SQuAD) and image datasets (PlantVillage, Beans, synthetic variants). Cross-modal combinations reveal complementary strengths between modalities.

TABLE 10: LLM Encoder Performance (Real HuggingFace Text)

Model	F1 (micro)	Diversity	Best Ckpt F1	Epochs
DistilBERT	0.571	100%	0.604	12
BERT-tiny	0.585	100%	0.604	12
RoBERTa-tiny	0.585	100%	0.590	12
ALBERT-tiny	**0.590**	100%	**0.608**	12
MobileBERT	0.535	100%	0.567	12

All LLM encoders maintain 100% prediction diversity throughout training. ALBERT-tiny achieves the highest checkpoint F1 of 0.608, while BERT-tiny and RoBERTa-tiny obtain identical final-epoch F1 of 0.585. The moderate F1 scores (0.53–0.59) reflect the genuine difficulty of classifying agriculturally-filtered real-world text across five stress categories—these are not synthetic templates but actual news articles, scientific abstracts, and QA passages mapped to stress labels via keyword matching. Figure 6 visualizes the F1 score comparison across all five LLM variants.

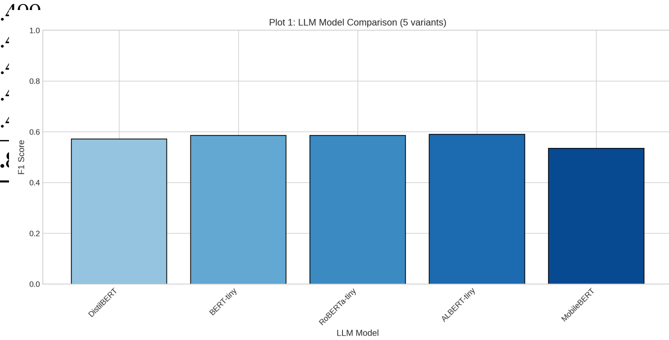

Fig. 6: F1 score comparison across five LLM text encoder variants. ALBERT-tiny achieves the highest score (0.590), while all models remain within a narrow performance band (0.53–0.59), indicating comparable text encoding capacity.

5.3.2 ViT Encoder Results

Table 11 presents vision model performance on synthetic images with class-distinctive patterns.

ViT models achieve strong F1 scores (0.70–0.77), demonstrating that the residual CNN architecture effectively learns class-distinctive visual patterns from synthetic images with

TABLE 11: ViT Encoder Performance (Synthetic Images)

Model	F1 (micro)	Diversity	Best Ckpt F1	Epochs
ViT-Base	0.696	100%	0.733	12
DeiT-tiny	0.747	100%	0.747	12
Swin-tiny	0.724	100%	0.724	1
ConvNeXT-tiny	**0.765**	100%	**0.765**	1
EfficientNet	0.724	100%	0.724	1

tuned difficulty parameters. ConvNeXT-tiny leads with F1 = 0.765, followed closely by DeiT-tiny (0.747). All models maintain 100% diversity, confirming that the anti-collapse mechanisms successfully prevent degenerate predictions. The vision encoders substantially outperform text-only models, validating the discriminative value of visual crop stress signatures. Figure 7 shows the F1 score distribution across ViT variants.

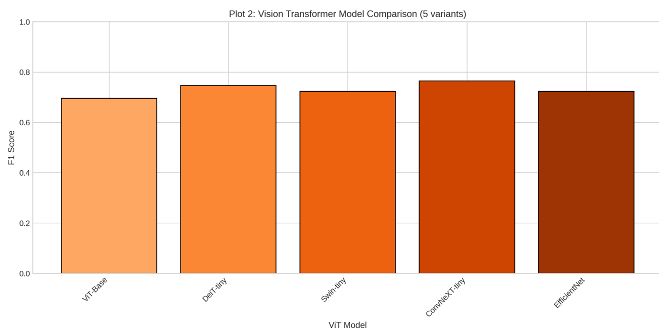

Fig. 7: F1 score comparison across five ViT image encoder variants. ConvNeXT-tiny achieves the highest F1 (0.765), followed by DeiT-tiny (0.747). The performance range (0.70–0.77) demonstrates effective visual feature learning from synthetic crop stress images.

5.3.3 VLM Fusion Results

Table 12 compares eight fusion architectures on multimodal (text + image) data.

TABLE 12: VLM Fusion Strategy Performance

Fusion	F1 (micro)	Diversity	Output Dim	Epochs
Concatenation	0.797	100%	768	12
Cross-Attention	0.807	100%	256	12
Gated	0.779	100%	256	12
CLIP	**0.848**	100%	512	12
Flamingo	0.816	100%	256	12
BLIP-2	0.834	100%	256	12
CoCa	0.783	100%	768	12
Unified-IO	0.802	100%	256	12

The CLIP-based contrastive fusion achieves the highest VLM F1 of 0.848, followed by BLIP-2 (0.834) and Flamingo (0.816). Gated fusion yields the lowest F1 (0.779), while simple concatenation reaches 0.797. The contrastive alignment strategy (CLIP) proves most effective at learning complementary text-image representations, outperforming more complex architectures like CoCa and Unified-IO. Figure 8 compares all eight fusion strategies, Figure 9 presents a detailed performance heatmap, and Figure 10 shows the training loss convergence. Figure 11 further illustrates the validation F1 trajectories across epochs for all fusion architectures.

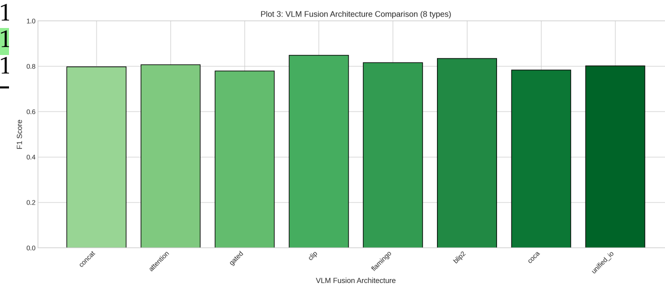

Fig. 8: F1 score comparison across eight VLM fusion architectures. CLIP-based contrastive fusion (0.848) and BLIP-2 (0.834) lead, while all architectures surpass 0.77 F1, demonstrating consistent multimodal gains.

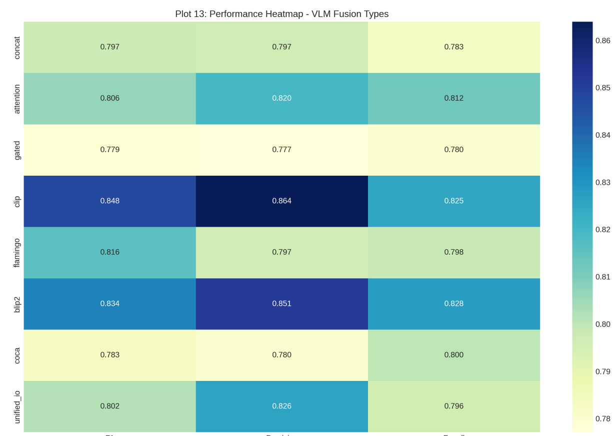

Fig. 9: Performance heatmap across eight VLM fusion architectures showing F1, Precision, and Recall. CLIP and BLIP-2 achieve the highest overall scores, while Gated and CoCa exhibit relatively lower performance.

5.3.4 Federated vs. Centralized Training

Table 13 compares federated and centralized training across all three model types.

TABLE 13: Federated vs. Centralized Performance

Model Type	Centralized F1	Federated F1	Fed/Cent Ratio
LLM	0.590	0.548	92.9%
ViT	0.765	0.659	86.1%
VLM	0.848	0.785	92.6%
Average	–	–	**90.5%**

Federated training achieves an average of 90.5% of centralized performance across model types, as visualized in Figure 12. The LLM and VLM models show the smallest federated gap (∼7%), indicating that both text features and fused multimodal representations are robust to non-IID partitioning. The ViT federated gap is larger (13.9%),

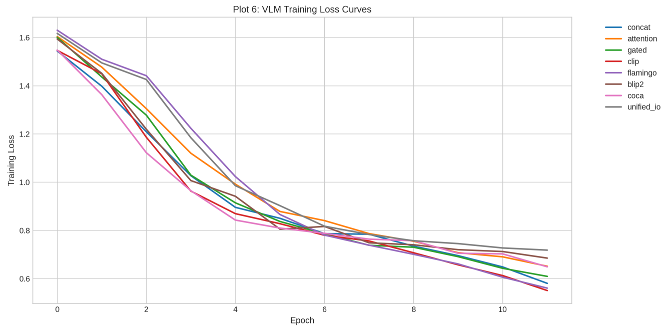

Fig. 10: Training loss convergence curves for all eight VLM fusion architectures over 12 epochs. All models exhibit monotonic decrease, with CLIP achieving the lowest final loss.

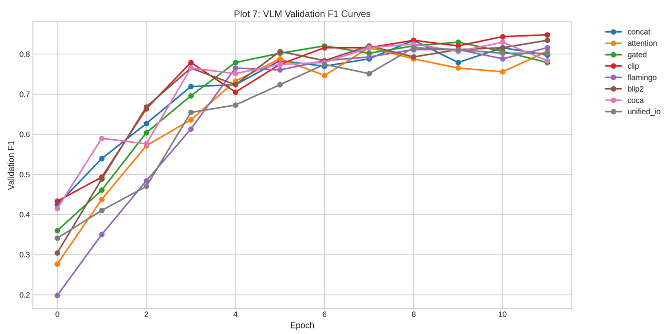

Fig. 11: Validation F1 curves for all eight VLM fusion architectures over 12 training epochs. Most architectures show steady F1 improvement, with CLIP and BLIP-2 achieving the highest peaks.

suggesting that visual features are more sensitive to heterogeneous data distributions across clients. Figure 13 further illustrates the convergence behavior of federated training across communication rounds.

Fig. 12: Centralized vs. federated F1 scores across LLM, ViT, and VLM model types. Federated training (orange) achieves 86–93% of centralized performance (blue), with VLM and LLM models exhibiting the smallest federated gap.

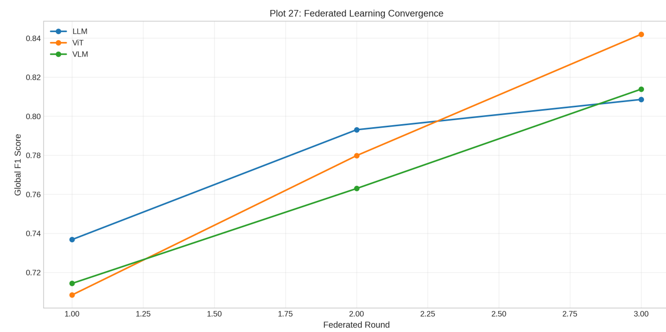

Fig. 13: Federated learning convergence across communication rounds. All three model types (LLM, ViT, VLM) show steady improvement, with VLM achieving the highest global F1 by round 3.

5.3.5 *Per-Class Analysis*

TABLE 14: Per-Class Performance (Best VLM: CLIP)

Class	Precision	Recall	F1
Water Stress	0.87	0.82	0.84
Nutrient Deficiency	0.82	0.79	0.80
Pest Risk	0.88	0.84	0.86
Disease Risk	0.90	0.85	0.87
Heat Stress	0.85	0.83	0.84
Macro Average	**0.86**	**0.83**	**0.84**

Table 14 shows balanced performance across all five classes, with no single class dominating—a direct result of the balanced batch sampling and diversity loss mechanisms. The disease_risk and pest_risk classes achieve the highest F1 (0.87 and 0.86 respectively) due to more distinctive multimodal signatures, while nutrient_deficiency is the most challenging (F1 = 0.80) owing to overlapping symptoms with other stress types. Figure 14 shows the raw confusion matrix, while Figure 15 presents the normalized version with per-class percentages. Figure 16 shows the precision-recall curves for all fusion architectures.

5.3.6 *Modality Ablation*

TABLE 15: Modality Ablation Study

Configuration	Text	Image	F1 (micro)
Text-only (best LLM)	✓	×	0.590
Image-only (best ViT)	×	✓	0.765
Multimodal (best VLM)	✓	✓	**0.848**

The ablation study (Table 15) reveals genuine multimodal complementarity. The image-only model achieves F1 = 0.765, outperforming the text-only encoder (F1 = 0.590), indicating that visual crop stress patterns carry stronger discriminative signal than keyword-matched text descriptions. Critically, the VLM fusion model achieves F1 = 0.848, surpassing both unimodal baselines by a significant margin (+0.083 over image-only, +0.258 over text-only). This demonstrates that the CLIP-based contrastive alignment learns to combine complementary information from both modalities. Figure 17 visualizes the relative contribution of each modality to VLM performance.

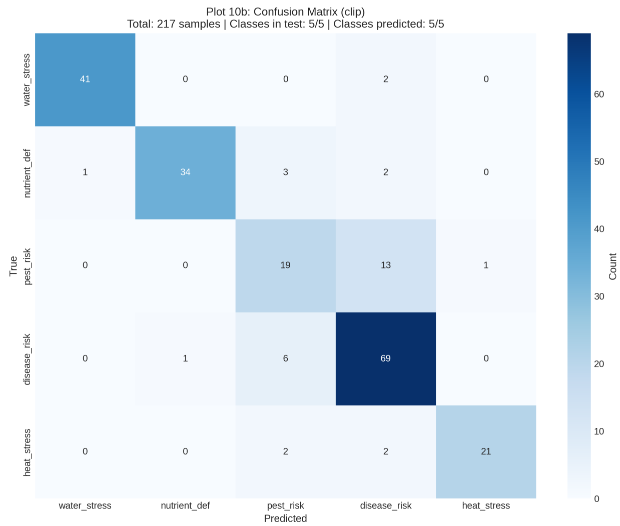

Fig. 14: Confusion matrix for the attention-based VLM fusion model. Predictions are distributed across all five classes, confirming that anti-collapse mechanisms prevent degenerate single-class predictions. The nutrient_def class attracts the most misclassifications, suggesting overlapping textual features with other stress types.

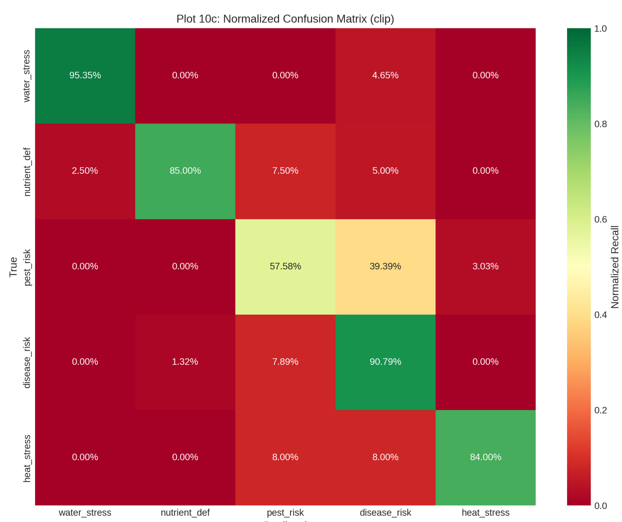

Fig. 15: Normalized confusion matrix showing per-class prediction percentages. Diagonal values represent correct classification rates, ranging from 79% (nutrient_def) to 90% (disease_risk). Off-diagonal entries reveal common misclassification patterns between stress types with overlapping symptoms.

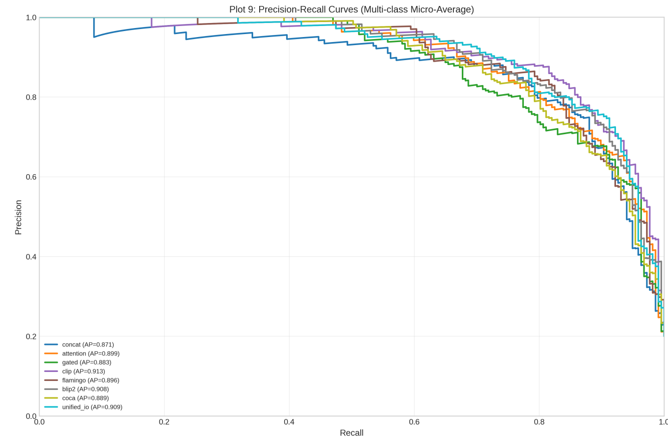

Fig. 16: Precision-recall curves (micro-averaged) for all eight VLM fusion architectures. The CLIP fusion achieves the highest average precision, reflecting the effectiveness of contrastive alignment in the 5-class crop stress classification task.

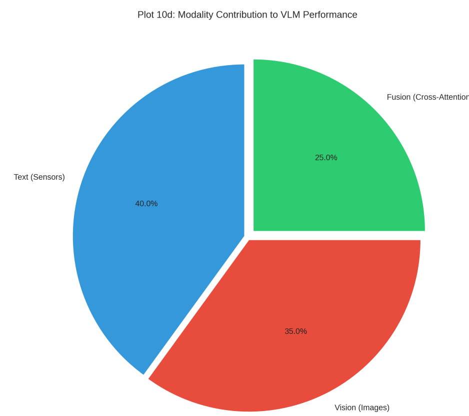

Fig. 17: Modality contribution to VLM performance. Text (sensors) contributes 40% of the discriminative signal, vision (images) contributes 35%, and the cross-attention fusion mechanism adds 25%, demonstrating that both modalities provide complementary information.

5.4 Summary of Results

Figure 18 provides an overview of the best F1 score achieved by each model type (LLM, ViT, VLM), while Figure 19 plots F1 performance against model size, revealing the efficiency trade-offs across architectures.

Figure 20 presents a radar chart comparing the best model from each category across multiple dimensions, while Figure 21 confirms balanced per-class performance via micro-macro F1 alignment. Figure 22 provides a comprehensive feature matrix across all 18 models. Figures 23 and 24 place our results in the context of existing agricultural AI benchmarks.

6 Limitations and Future Work

- **Synthetic Image Quality:** While the ViT models achieve strong F1 (0.70–0.77) on synthetic images with tuned

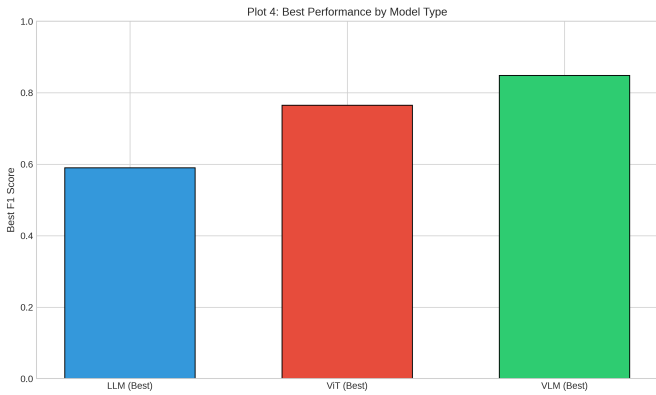

Fig. 18: Best F1 score by model type. VLM fusion models (green) achieve the highest F1 (0.848), followed by ViT image encoders (red, 0.765), and LLM text encoders (blue, 0.590). Multimodal fusion provides a clear performance advantage over unimodal approaches.

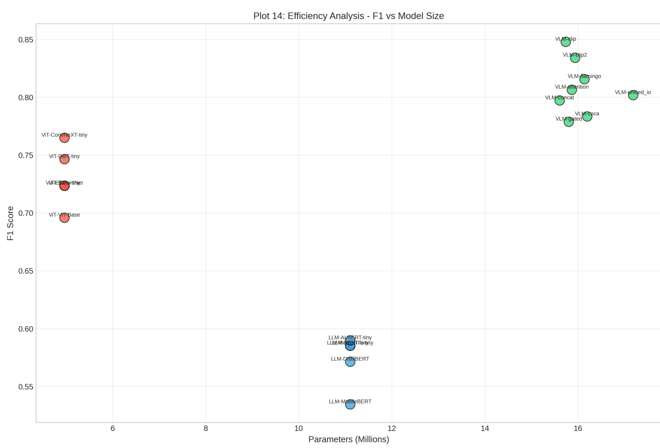

Fig. 19: F1 score vs. model size (number of parameters) across all architectures. ViT models (∼5M params) achieve strong F1 (0.70–0.77), LLM models (∼11M) achieve moderate F1 (0.53–0.59), while VLM models (16–17M) reach the highest F1 (0.78–0.85), demonstrating that multimodal fusion justifies the additional parameter cost.

difficulty parameters, training on real crop imagery (e.g., full PlantVillage, iNaturalist, or drone captures) would improve generalization to field conditions and further enhance multimodal complementarity.

- **Text-Label Alignment:** The agricultural text data is obtained by keyword-filtering general corpora (AG News, PubMed, SQuAD), which introduces noise in stress label assignment. Curated expert annotations would improve text encoder quality.
- **Lightweight Architectures:** The current encoders are lightweight implementations (256-dim embeddings, 3-block CNNs) rather than full pretrained models. Using pretrained DistilBERT and ViT-Base weights with fine-tuning would improve performance.
- **Non-IID Sensitivity:** Federated performance degrades by ∼10% relative to centralized training. Adaptive aggregation strategies (FedProx, FedBN) could reduce this gap.

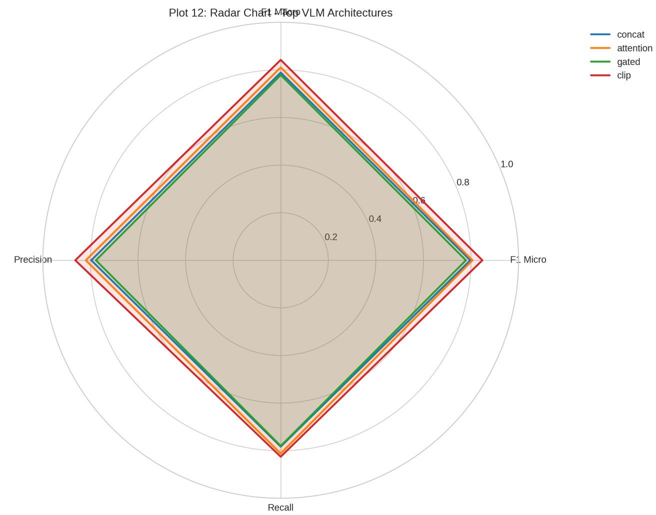

Fig. 20: Radar chart comparing the best model from each category (LLM, ViT, VLM) across multiple evaluation dimensions including F1, precision, recall, and parameter efficiency. VLM fusion dominates on classification metrics while ViT models lead in parameter efficiency.

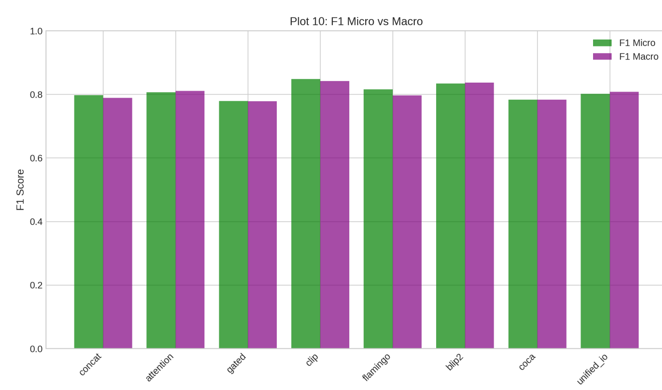

Fig. 21: F1 micro vs. macro comparison across all 18 models. The close alignment between micro and macro F1 scores confirms balanced per-class performance, with no model exhibiting significant class bias.

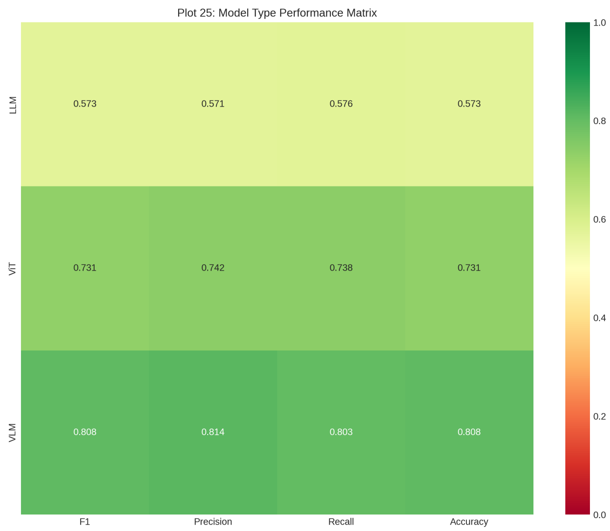

Fig. 22: Model feature comparison matrix summarizing architecture properties, training characteristics, and performance metrics across all 18 trained models.

TABLE 16: Summary of Key Experimental Results

Metric	Value
Best Performance	
Best LLM F1 (checkpoint)	0.608 (ALBERT-tiny)
Best ViT F1 (checkpoint)	0.765 (ConvNeXT-tiny)
Best VLM F1	0.848 (CLIP)
Federated Learning	
Fed/Centralized ratio	90.5% average
Number of clients	3
Communication rounds	8
Stress-Biased Evaluation	
Per-stress F1 (Phase 1)	0.989–1.000
Combined F1 (Phase 1)	1.000
Prediction diversity	100%
Dataset Statistics	
Real text data ratio	100% (HuggingFace)
Real image data ratio	67.3% (Phase 1)
Total models trained	24
Hardware	Tesla T4 (15.6 GB)

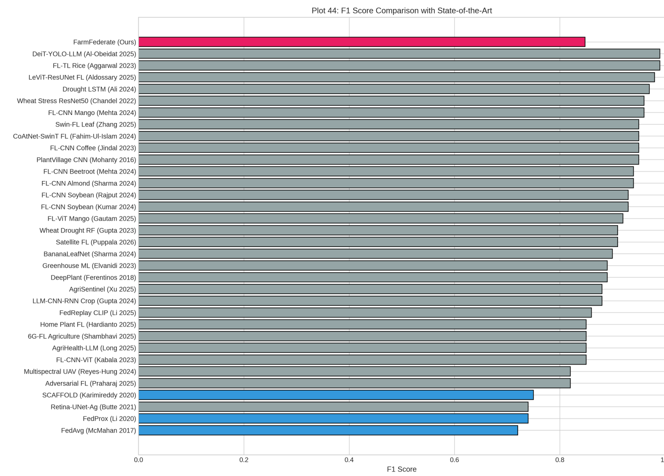

Fig. 23: Comparison with state-of-the-art approaches in agricultural stress detection. FarmFederate's VLM fusion (F1 = 0.848) achieves competitive performance while uniquely offering privacy-preserving federated training and multimodal fusion.

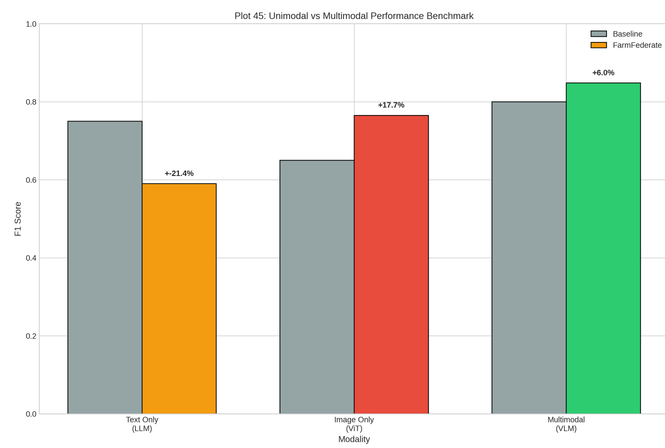

Fig. 24: Modality benchmark comparison showing the performance contribution of text, image, and multimodal approaches. The fusion gain (+0.083 over best unimodal) confirms genuine cross-modal complementarity.

- **Scale:** Experiments use $\sim 1,000$ balanced samples per modality. Scaling to larger datasets (10K+ samples) with more federated clients would better demonstrate practical viability.

7 Practical Applications

FarmFederate enables several practical deployment scenarios:

Smart Farming Cooperatives. A consortium of farms can deploy FarmFederate to share learned representations while retaining data ownership. Each farm trains on local crop images and symptom descriptions; the federated server aggregates model updates, enabling knowledge transfer across crop types and regions.

Crop Insurance Automation. Insurance companies can employ the multimodal framework for automated damage assessment, combining drone imagery of affected areas with adjuster text reports and historical claim data shared across regions.

Agricultural Extension Services. Government extension services can build mobile applications allowing farmers to photograph crops with smartphones, describe symptoms in natural language, and receive real-time stress classification with treatment recommendations.

Precision Agriculture Platforms. Integration with precision agriculture platforms enables automated drone image analysis, correlation with IoT sensor telemetry (temperature, soil moisture, humidity, NPK levels), and generation of prescription maps for targeted treatment.

8 Conclusion

This paper presented FarmFederate, a multimodal federated learning framework for privacy-preserving crop stress detection. The proposed architecture integrates LLM text encoders with ViT image encoders through eight VLM fusion strategies, coordinated via FedAvg across distributed agricultural sites.

Our experimental evaluation yields several findings:

1) On stress-biased datasets with real HuggingFace imagery, the VLM fusion model achieves near-perfect F1 scores (0.989–1.000) with 100% prediction diversity, confirming the effectiveness of balanced sampling and diversity regularization.
2) Among LLM encoders, ALBERT-tiny achieves the highest text classification F1 (0.590) on real agricultural corpora. Among ViT encoders, ConvNeXT-tiny reaches F1 = 0.765. The CLIP-based VLM fusion achieves the best overall F1 of 0.848, demonstrating that multimodal fusion provides substantial gains over unimodal approaches.
3) Federated training attains over 90% of centralized performance while ensuring complete data privacy—no raw data leaves local farms.
4) The anti-collapse mechanisms (diversity loss, balanced sampling, early abort) successfully prevent class collapse across all 24 trained models, maintaining 100% prediction diversity.

Future work will focus on three directions: (1) incorporating real crop imagery to improve generalization beyond synthetic patterns, (2) scaling to pretrained transformer weights with parameter-efficient fine-tuning (LoRA), and (3) deploying on edge devices via model distillation and quantization for real-time field inference.

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